THE TACTICAL EVOLUTION OF EUROPEAN FOOTBALL (2013-2025)

A Data-Driven Analysis of Formation Trends Across Top 5 Leagues

Access the GitHub repo here: https://github.com/philippgeiger/AM10_AUT25_group_10

GROUP 10

Sarah Alvi, Ediz Arslan, Kinson Cheung, Philipp Geiger, Neha Naeem, Amy Su

Key Questions

1	Which formations perform best?
2	Is there an evolution in favoured formations?
3	Do budgets change tactical choices?
4	Does cross-league contagion exist?
5	Do team formations change depending on where they play?

Background & Preparation

Our Sources

- Data set of **21,276 matches**
- Leagues we analysed:
  - Premier League
  - La Liga
  - Serie A
  - Bundesliga
  - Ligue 1
- Seasons: 2013-2014 to 2024-2025

How we Prepared

- SQL filters to isolate Big 5 leagues and join games, club_games, clubs, lineups
- Built club total_market_value by summing player values
- Parsed formation strings to count defenders
- Used Python (pandas, seaborn, matplotlib, geopandas) and R for cleaning and visualisation
- Consulted tactical diffusion literature to frame cross-league comparisons

Methodology & Statistical Techniques

Core Approach

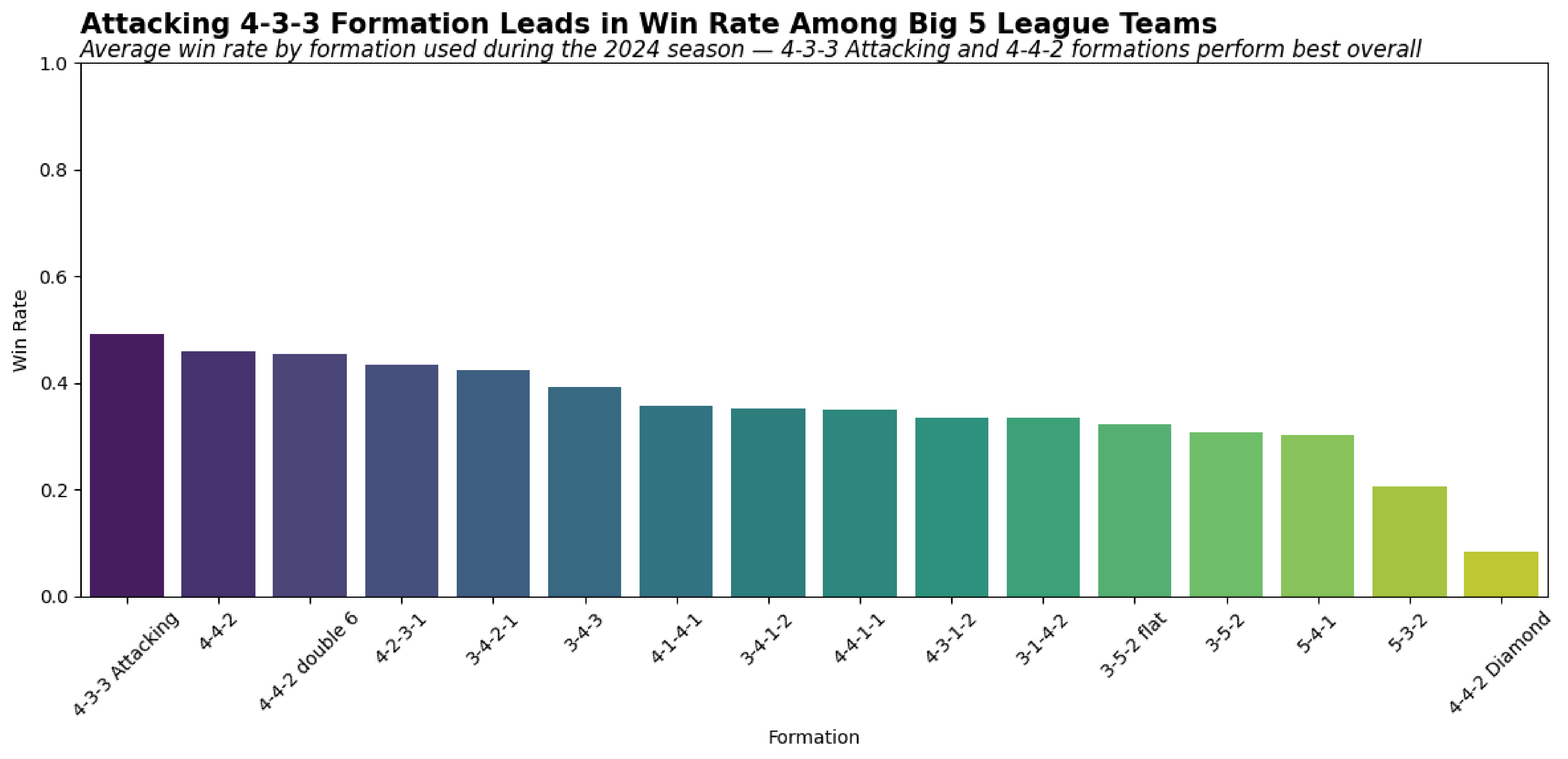
- Descriptive statistics: counts, percentages, averages, Pearson Correlation
- Controlled for context by analysing Home and Away separately, before averaging
- Reliability ensured by requiring minimum sample sizes (for example a minimum of 10-20 games per formation)

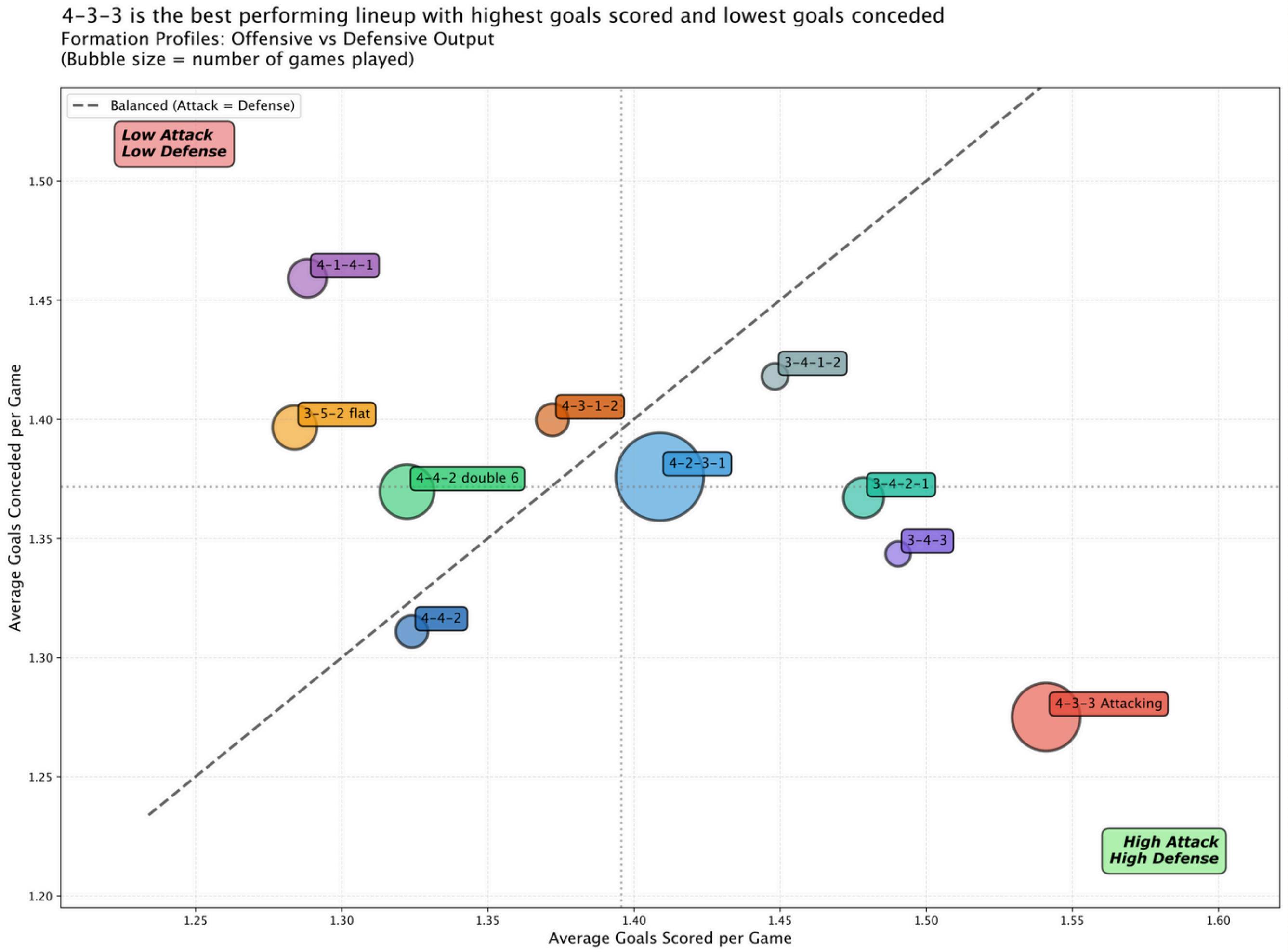
Techniques

- Bar charts - win rate by formation
- Line charts - formation usage over time + defenders per game
- Scatterplots - attack vs defence + home vs away risk-reward
- Correlation heatmaps - cross-league tactical contagion
- Geographic mapping - dominant formation and share by country

Which Formations Perform Best?

Attacking 4-3-3 Formation Takes the Win





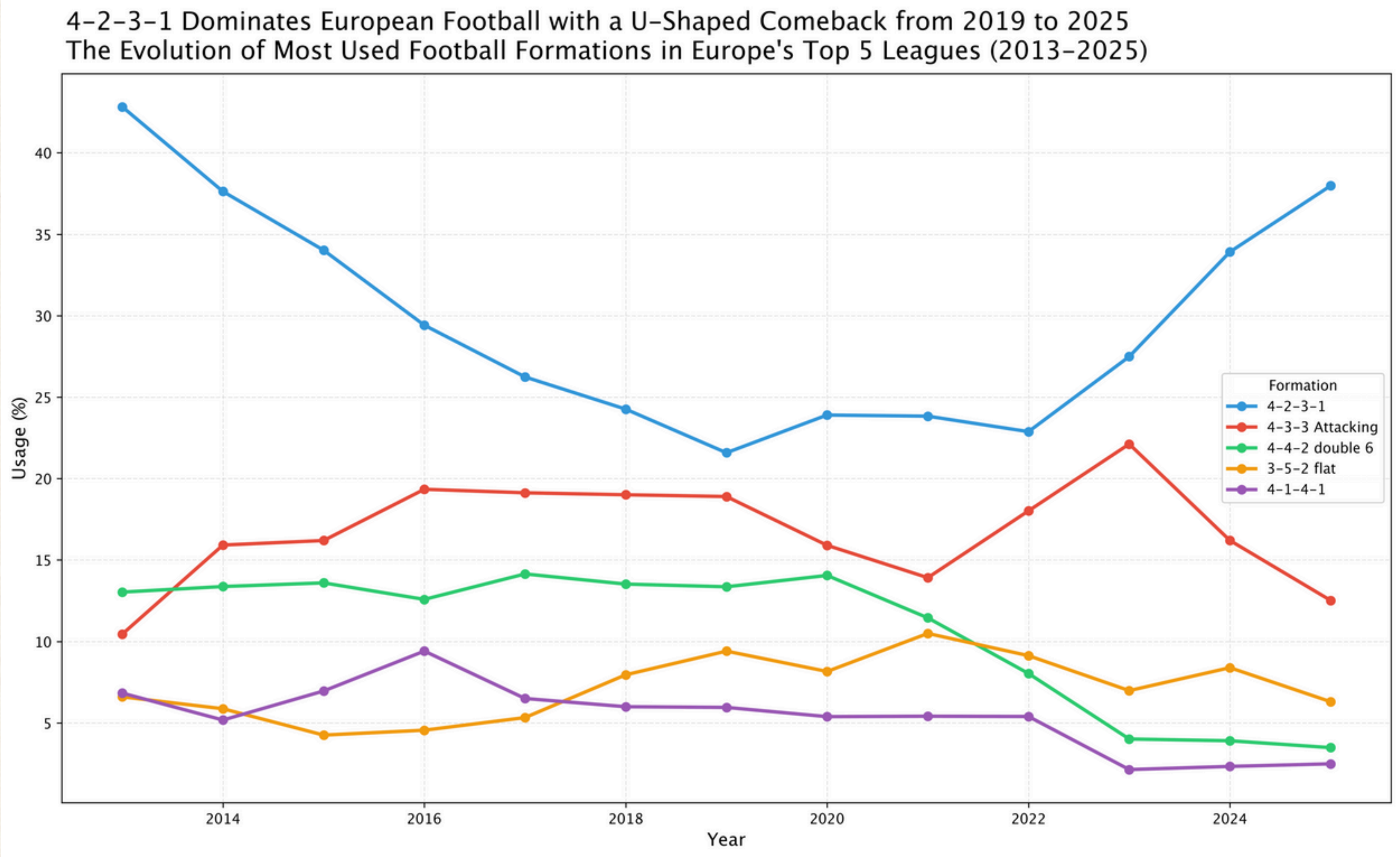
Built from SQL joins of club_games and games; grouped by formation, filtered to ≥10 matches. Used Seaborn and Matplotlib to create a bar chart showing the win rate for each formation:
The Attacking 4-3-3 leads in win rate, 4-4-2 and 4-2-3-1 are also strong, while defensive systems like 5-3-2 and 4-4-2 Diamond have lower win rates.

Using formations with ≥20 home and away games data points, we averaged goals scored and conceded across venues; the bubble size = sample size.

Bottom-right is ideal. 4-3-3 Attacking and 4-2-3-1 cluster as net-positive and have high-output shapes.

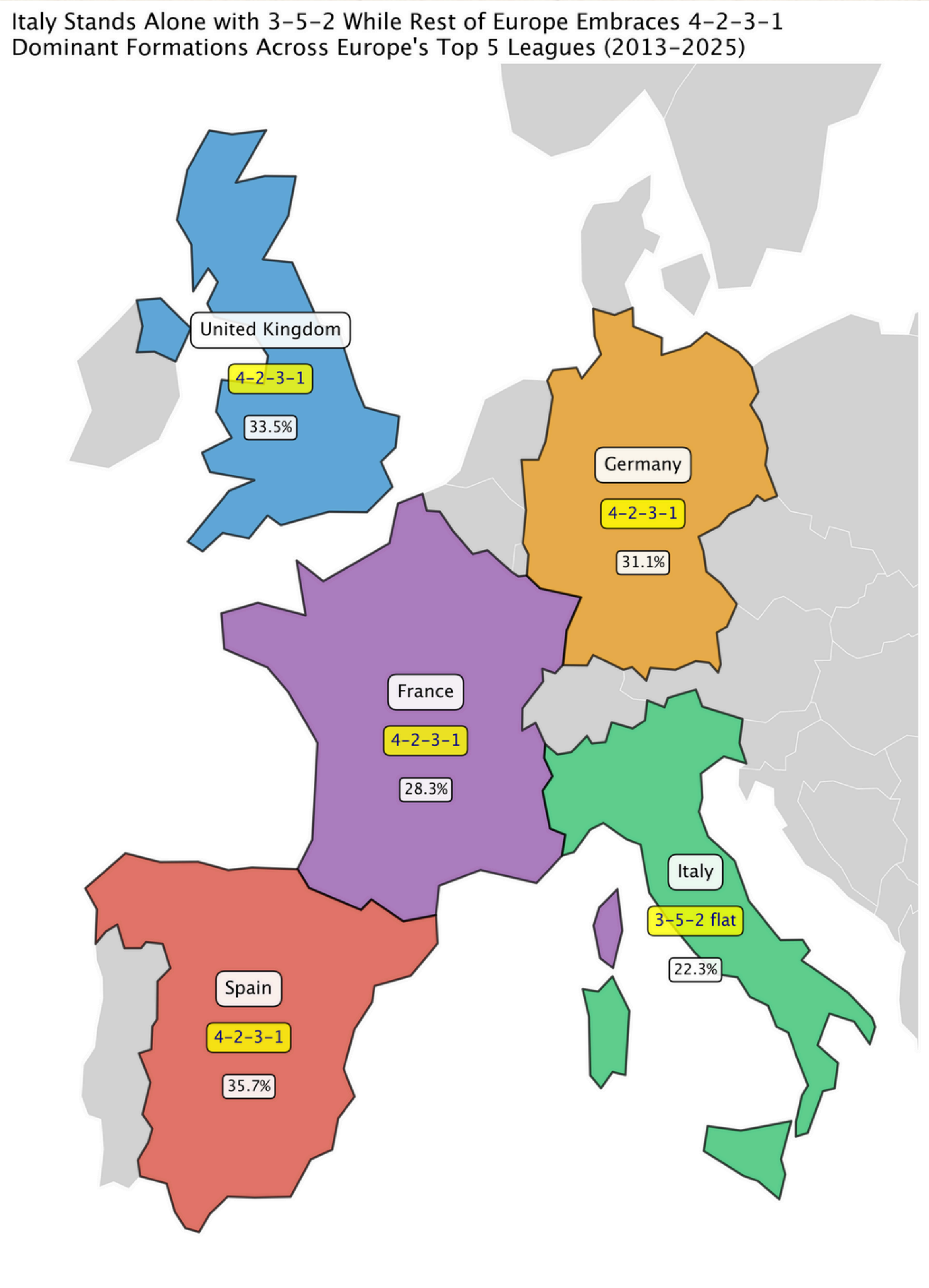
Evolution Over Time & Geographic Identity

4-2-3-1 Makes a Dramatic U-Turn



For each season we converted formation counts into usage percentages. This graph shows how formation preferences have changed over time.

4-2-3-1 shows a pronounced U-shape which is dominant in 2013, declining in the late 2010s, then re-emerging as the leading form by 2025.

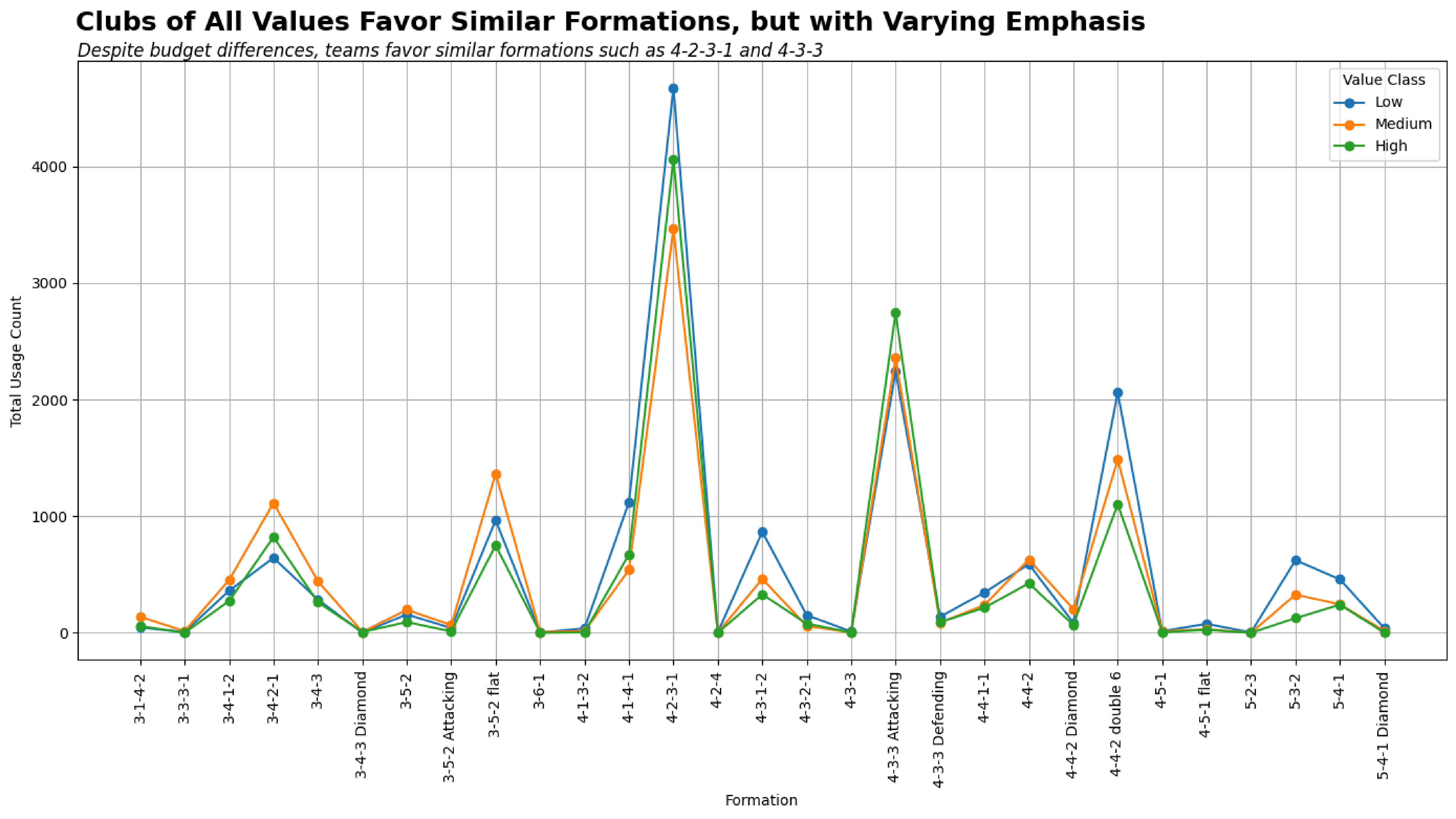


Using geopandas to pull map data for European countries, we combined 13 years of data per league identifying the single most-used formation and its share.

England, Spain, Germany and France all favour 4-2-3-1 (~28–36%), while Italy stands alone with 3-5-2 at 22.3%, underlining a distinct tactical culture.

Do Budgets Change Tactical Choices?

Budgets Change Emphasis, Not Formation Choice



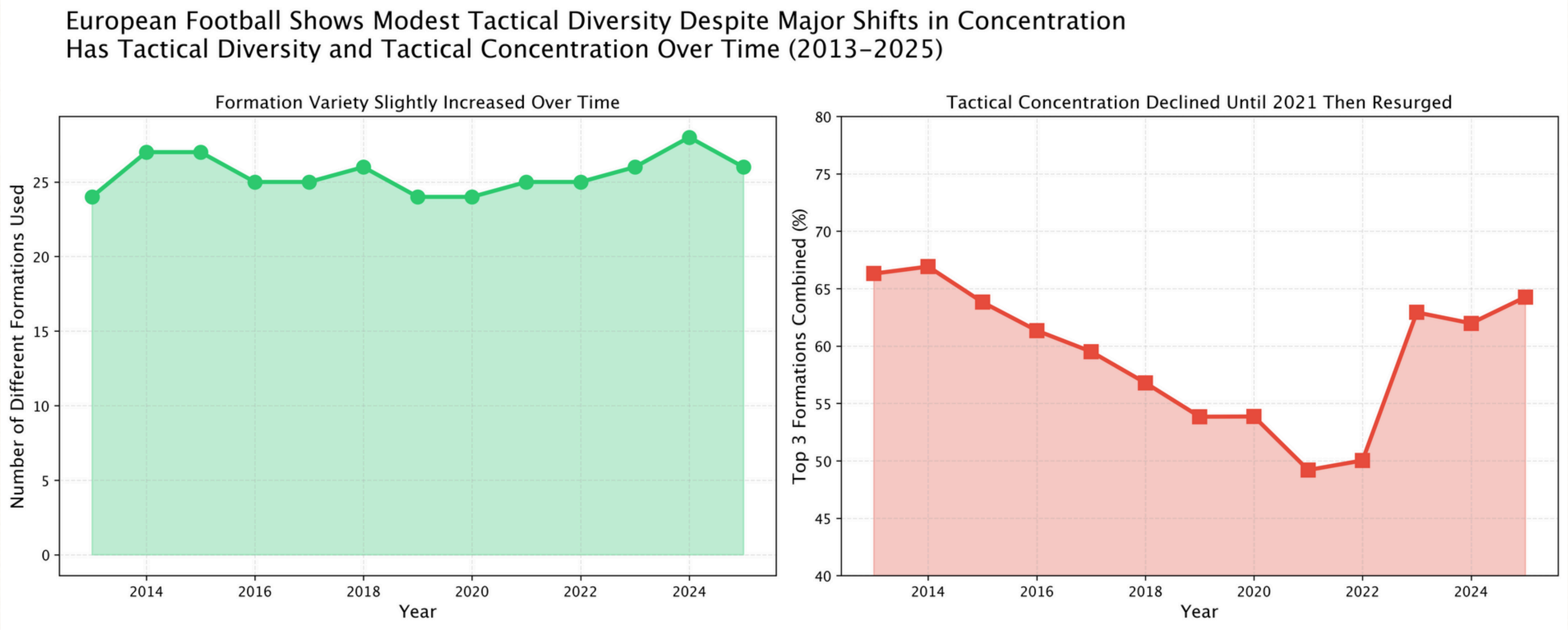
Since clubs didn't have a total value in the dataset, we computed each club's total_market_value by summing the market value of all of its players.

Using summed player values, clubs were classified by budget:
Low: €0–150M
Medium: €150–400M
High: €400M+

All budget tiers favour similar formations, especially 4-2-3-1 and 4-3-3. Richer clubs use these dominant systems more often, but lower-budget clubs follow a similar pattern at a smaller scale.

Tactical Diversity & Cross-League Contagion

Leagues Copy Each Other's Successful Tactics

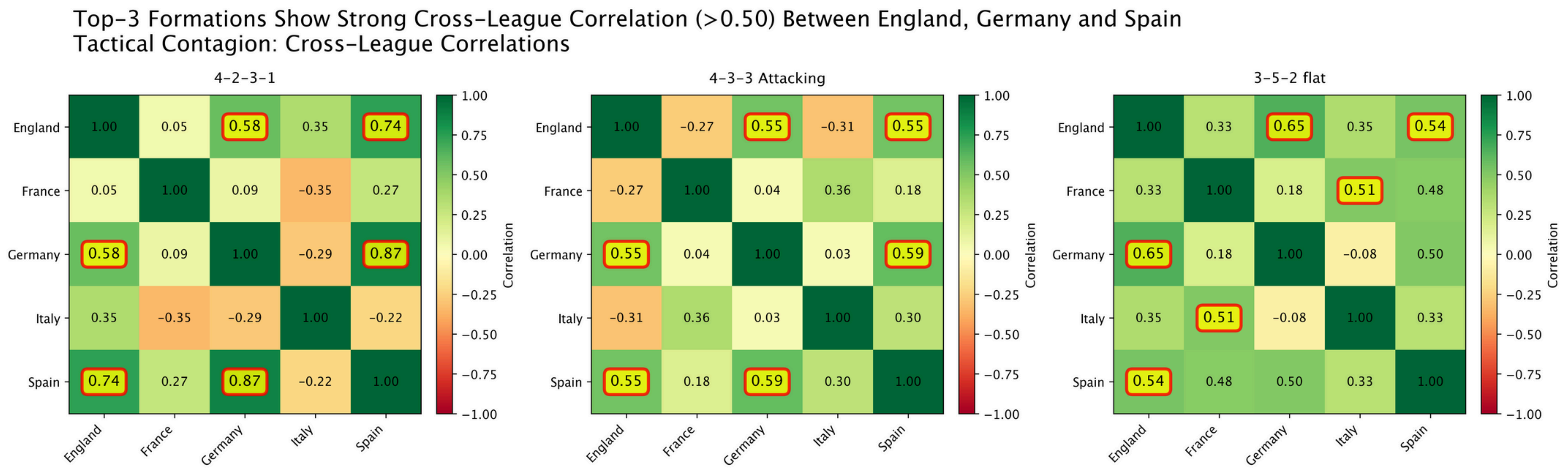


Yearly counts of distinct formations (left) show slight growth, but the share taken by the top three formations (right) also rises. These panels together answer whether European football is becoming more diverse.

We concluded that a paradox exists: European football offers more options, yet usage is increasingly concentrated around a few formations.

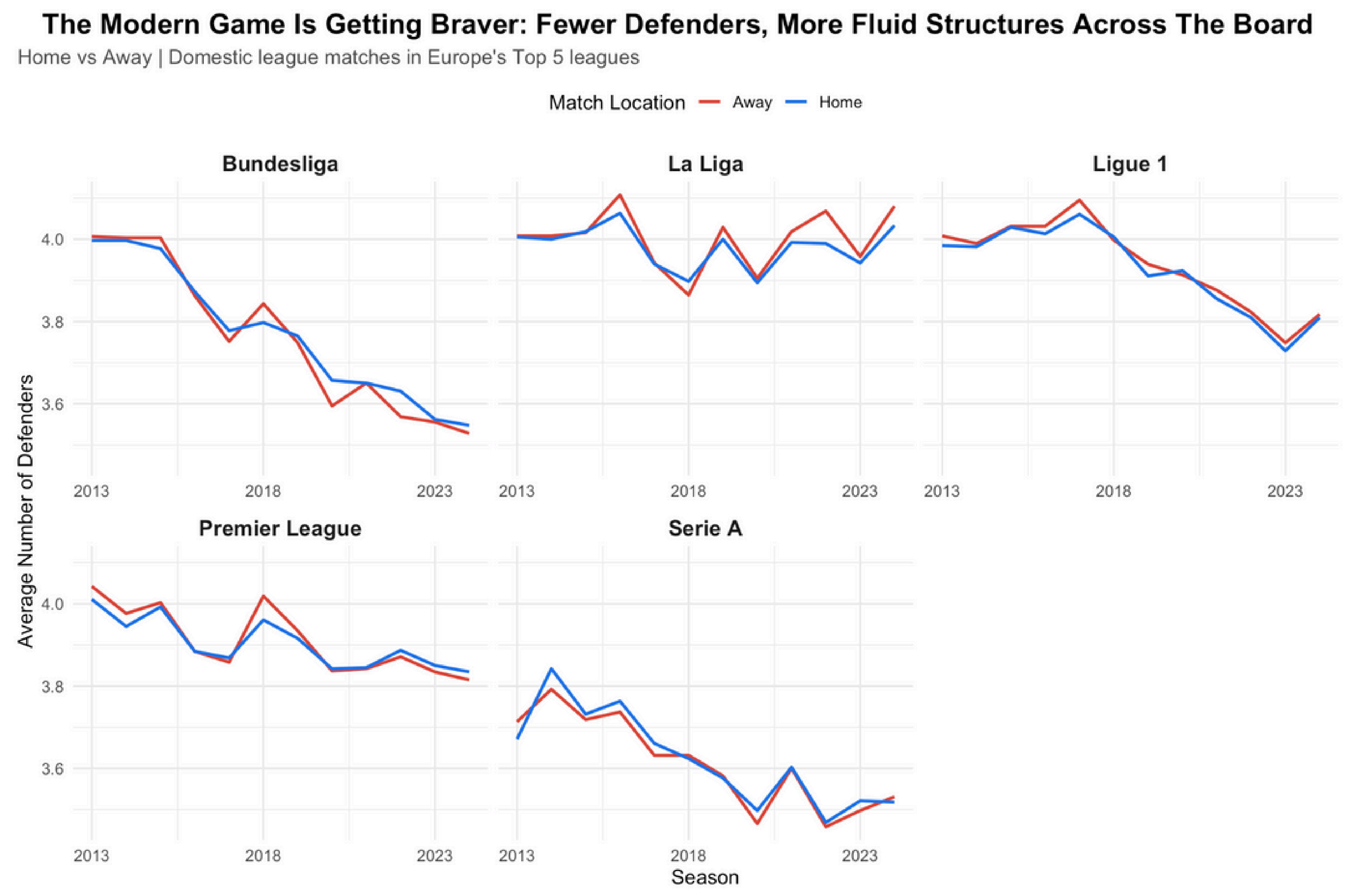
For each key formation we built league-by-year usage series and computed Pearson correlations. Values above 0.50 (highlighted) reveal strong synchronisation, especially for 4-2-3-1 between England, Germany and Spain.

In contrast, 3-5-2 shows weaker links, supporting Italy's tactical independence.



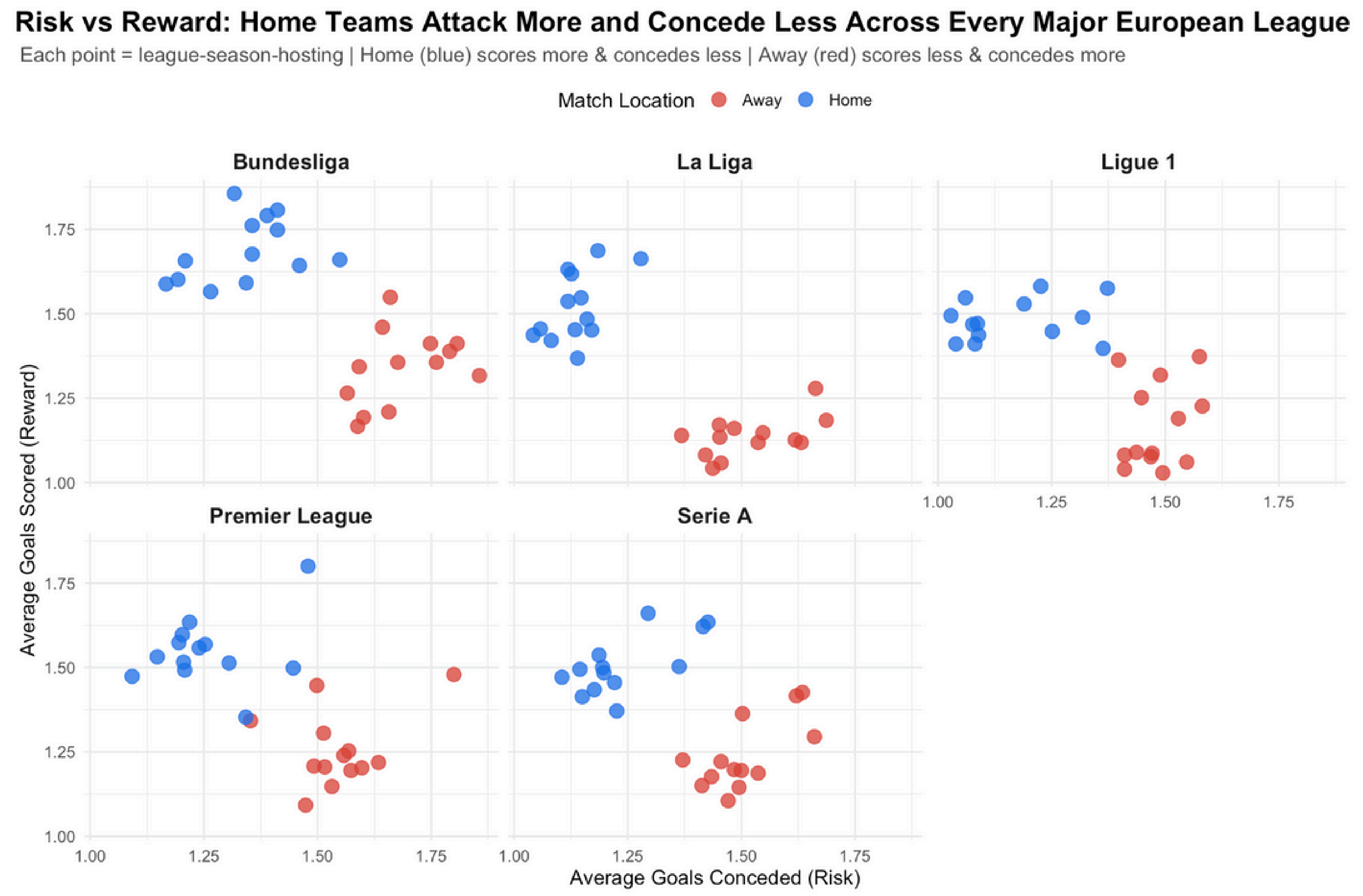
Home vs Away: Structure & Risk-Reward

A Stronger Attack Requires Less Defence - High Risk, High Reward



R parsed the first number in each formation (e.g. “4-2-3-1” → 4) and averaged defenders by league, season and venue.

Across all leagues, the average number of defenders drifts downward, and away teams are not simply adding extra defenders. This supports the idea of more fluid, shared defensive responsibilities.



Aggregating goals scored and conceded by league-season-venue, each point is a home or away profile.

Home teams sit higher (more goals scored) and to the left (fewer goals conceded), confirming that home sides consistently take on more risk with higher reward.

Conclusions, Critique & Future Work

Conclusions

1
4-3-3 Attacking and 4-2-3-1 deliver the best balance of win rate and goal difference

2
Teams with different budgets still use the same formations

3
4-2-3-1's U-shaped comeback and cross-league correlations suggest tactical contagion

4
Italy’s 3-5-2 tradition remains a clear outlier

5
Home teams attack more and concede less; modern systems use fewer fixed defenders

Critique and Advice

- Formation labels and roles are imperfect proxies for tactics
- Our analysis is descriptive, not causal; regression and causal models would be natural next steps
- Future students should
  - Plan for sample-size filters early
  - Separate home/away effects
  - Combine visualisations with simple, clearly explained statistics